OPEN ELECTIVE · SEMESTER 6 · 60 PERIODS

Contemporary Electronics

A full-screen handbook for Course Code **6042B**. This guide converts the syllabus into a complete learning document covering daily-life electronics, entertainment devices, domestic appliances, backup systems, batteries, electric vehicles, charging methods, connectors, standards, calculations, diagrams, revision points and model answers.

PROGRAM
Electronics / E&C Engineering

CREDITS
4

PERIODS
4 per week / 60 per semester

LEVEL
Understanding

□ □□□□□□□□□ theory □□□□□□ □□□□. □□□ device-□□□ construction, working principle, block diagram, specification, maintenance, application □□□□□□ □□□□□□□ □□□□□□□□ answer strong □□□□.

SYLLABUS DECODED

Official Course Structure

The syllabus objective is to provide knowledge-level information about electronic devices used in everyday life. The course has four outcomes: applications of electronics, entertainment devices, domestic appliances and power backup, and EV charging standards.

CO	Outcome	Hours	Handbook Focus
CO1	Interpret the application of electronics in everyday life.	11	Evolution from vacuum tube to IC, Moore's Law, microprocessor vs microcontroller, consumer products.
CO2	Explain working principle of entertainment electronic devices.	14	Microphone, speaker, AM/FM receiver, stereo, home theatre, display, video ports and media players.
CO3	Explain domestic appliances and power backup devices.	17	Washing machine, microwave oven, induction stove, AC, solar system, UPS, inverter, battery ratings and maintenance.
CO4	Explain electric vehicle and charging standards.	16	EV schematic, charger block, slow/fast charger, AC/DC charging, EVSE modes, connectors and IEC standards.
Series Tests	Two internal tests	2	Use the revision checklists and model question paper.

How to write answers

1. Start with definition or purpose.
2. Draw a neat block diagram or simple schematic.
3. Explain each block in the signal/power flow order.
4. Add specifications, advantages, limitations and applications.
5. Finish with maintenance/safety if the device is appliance, UPS, battery or EV charger.

Engineering connection

For an electronics technician, this subject is practical: panel displays, SMPS, UPS backup, sensors, motor-driven appliances, display interfaces and EV charging are common field-service areas.

> Block diagram □□□□□□□□□□□□□□□□□ fault finding □□□□□□□□□□□□. Power path, signal path, control path □□□□□□□□□□□□ □□□□□□.

MODULE 1 · CO1 · 11 HOURS

Consumer Electronics Fundamentals

History of electronic devices, vacuum tubes, transistors, integrated circuits, Moore's Law, microprocessors, microcontrollers and their use in consumer electronics.

Evolution of electronic devices

Generation	Device	Main feature	Limitation
1	Vacuum tube	Controls electron flow in vacuum; used in early radios and computers.	Large size, heat, high power, less reliable.
2	Transistor	Small solid-state device for switching/amplification.	Discrete wiring becomes complex for large circuits.
3	Integrated Circuit	Many components fabricated on one chip.	Requires semiconductor manufacturing and careful packaging.
4	Microprocessor	CPU on a chip; needs external memory and I/O.	Not ideal alone for simple dedicated products.
5	Microcontroller	CPU + memory + I/O + timers/ADC in one chip.	Limited speed/memory compared with full computer.

Vacuum tube → transistor → IC → microprocessor → microcontroller □□□□□□□□ basic development path.

Moore's Law

Moore's Law is the observation that transistor count in ICs tends to increase rapidly with technology advancement. In consumer electronics it results in smaller, faster, cheaper and more power-efficient products.

Exam point: Do not write only “transistors double.” Add practical effect: better mobile phones, smart TVs, digital cameras, IoT devices and embedded controllers.

Microprocessor vs Microcontroller

✓ Microprocessor: CPU only; used where high processing is needed.

- ✓ Microcontroller: CPU, memory, I/O and peripherals on one chip.
- ✓ Microcontroller is preferred in washing machine, remote, AC, microwave and toys because cost and power are low.

Consumer controller blocks

Sensor/input → microcontroller → display/actuator is the most common control pattern in consumer electronics.

- ✓ Input: button, remote, sensor.
- ✓ Processing: firmware and timing logic.
- ✓ Output: display, buzzer, relay, motor driver.

Applications

- ✓ TV and set-top box decoding and display control.
- ✓ Washing machine sequence and motor control.
- ✓ Microwave timer, keypad, display and magnetron control.
- ✓ AC temperature and compressor control.
- ✓ UPS battery monitoring and switching.

MODULE 2 · CO2 · 14 HOURS

Entertainment Electronics Systems

Microphones, loudspeakers, AM/FM receivers, stereo, 2.1/5.1 home theatre, display systems, video resolution, ports and video players.

Audio entertainment chain

1. Microphone converts sound into electrical signal.
2. Pre-amplifier raises weak signal level.
3. Tone or digital processor shapes audio quality.

4. Power amplifier supplies enough power.
5. Loudspeaker converts electrical signal back to sound.

Microphone and loudspeaker

Item	Working principle	Key points
Microphone	Sound pressure vibrates diaphragm and produces electrical variation.	Dynamic, condenser and electret types are common.
Loudspeaker	Current in voice coil creates magnetic force, moving cone and producing sound.	Impedance, wattage and frequency response matter.
2.1 system	Two satellite speakers + one subwoofer.	Good for music and computer entertainment.
5.1 system	Five full-range channels + subwoofer.	Front L/R, center, surround L/R and subwoofer.

AM and FM receiver

1. Antenna receives radio frequency signal.
2. Tuner selects the required station frequency.
3. RF/IF stages amplify and filter the selected signal.
4. Detector/demodulator extracts audio.
5. Audio amplifier drives the loudspeaker.

> Radio receiver-□ antenna signal □□□□□□□□□□□□, tuner station select □□□□□□□□□□□, demodulator audio □□□□□□□□□□□□□□□□.

Display parameters

Term	Meaning	Practical effect
1080p	1920 × 1080 progressive scan.	Full HD standard.
2K	Approx. 2048 horizontal pixels.	Cinema/production reference.

Term	Meaning	Practical effect
UHD / 4K	3840 × 2160.	Four times 1080p pixel count.
8K	7680 × 4320.	Very high detail and bandwidth need.

Ports and connectors

VGA

Analog video port used in older monitors/projectors. Video only.

Composite AV

Yellow RCA for video and red/white for audio. Simple but lower quality.

S-Video

Separates brightness and colour information. Better than composite but mostly obsolete.

HDMI

Digital audio and video through one cable. Supports HD and 4K depending version.

Optical port

Uses light through fibre for digital audio and avoids electrical noise.

DVD / Blu-ray

Blu-ray supports higher storage and video quality than DVD.

Common mistake: HDMI normally carries both digital video and audio.

MODULE 3 · CO3 · 17 HOURS

Home Appliances, Solar, UPS and Batteries

Working principle of domestic appliances and power backup systems: washing machines, microwave oven, induction stove, air conditioners, solar systems, UPS/inverter and storage batteries.

Washing machine

1. User selects program using control panel.
2. Controller checks door lock and water level sensor.
3. Inlet valve fills water; motor rotates drum/agitator.
4. Drain pump removes water.
5. Spin cycle removes moisture using high-speed rotation.

Important parts: drum, motor, inlet valve, drain pump, pressure sensor, door lock and control board.

Microwave oven and induction stove

Device	Energy conversion	Working point
Microwave oven	Electrical → microwave → heat in food.	Magnetron produces microwave radiation that agitates water molecules.
Induction stove	Electrical → magnetic field → eddy current heat.	High-frequency current in coil heats ferromagnetic vessel.

Microwave water molecule heating □□□□□□□□□□□□□□□. Induction stove vessel-□ eddy current heating □□□□□□□□□□□□□□.

Air conditioner basics

AC removes indoor heat using vapour compression refrigeration. Main parts: compressor, condenser, expansion device, evaporator, fan, refrigerant and control board.

Tonnage estimate ≈ Room heat load / 12000 BTU per hour

- ✓ **Non-inverter AC:** compressor switches ON/OFF at fixed speed.
- ✓ **Inverter AC:** compressor speed varies according to load; better comfort and energy saving.
- ✓ **1 ton AC:** approximately 12000 BTU/hour cooling capacity.

Solar power system

On-grid: connected to utility grid and usually works without battery. **Off-grid:** independent system requiring battery backup.

1. PV panel converts sunlight into DC.
2. Charge controller regulates charging.
3. Battery stores energy in off-grid system.
4. Inverter converts DC to AC for loads.

UPS and inverter

- ✓ **Offline UPS:** load normally on mains; switches to battery inverter during failure.
- ✓ **Online UPS:** load continuously supplied through inverter; no transfer delay.
- ✓ **Line-interactive UPS:** includes voltage regulation and battery backup.
- ✓ **Inverter:** converts DC battery supply to AC for longer backup.

UPS VA rating ≈ Total load power in watts / Power factor

Backup time = Battery voltage × Ah × efficiency / Load watts

Storage batteries

Term	Meaning
Primary battery	Non-rechargeable cell.

Term	Meaning
Secondary battery	Rechargeable battery.
Ah rating	Current capacity over time.
Wh rating	Energy capacity = voltage × Ah.
VRLA	Valve Regulated Lead Acid sealed battery.

Battery safety: Avoid sparks, short circuits, reverse polarity, overcharging, poor ventilation and low electrolyte level.

MODULE 4 · CO4 · 16 HOURS

Electric Vehicle and Charging Standards

EV schematic, charging equipment, slow/fast chargers, AC/DC charging, EVSE modes, charging time calculation, connectors and communication standards.

Simple EV schematic

1. Charge port receives energy from EVSE.
2. On-board charger converts AC to DC during AC charging.
3. Battery pack and BMS store and supervise energy.
4. Inverter converts DC to variable AC for traction motor.
5. Motor drives vehicle wheels through drivetrain.

AC and DC charging

Charging type	Where conversion happens	Typical use
AC charging	On-board charger converts AC to DC.	Home, workplace, slow/normal charging.
DC charging	External charger converts AC to DC.	Fast public charging.

Charging type	Where conversion happens	Typical use
Slow charger	Lower power rating and longer time.	Overnight charging.
Fast charger	Higher power, communication and thermal control.	Short stop charging.

AC charging-□ vehicle-□□□□ on-board charger □□□ conversion □□□□□□□□□□□. DC fast charging-□ conversion charger station-□ □□□□□□□□□□.

EVSE modes

- ✓ **Mode 2:** AC charging using cable with in-cable control/protection device.
- ✓ **Mode 3:** Dedicated AC charging station/EVSE with control pilot communication.
- ✓ **Mode 4:** DC fast charging with external charger and communication.

$$\text{Charging time} \approx \text{Battery capacity (kWh)} / \text{Charger power (kW)}$$

$$\text{Practical charging time} \approx \text{Battery capacity} / (\text{Charger power} \times \text{efficiency})$$

Connector selection

Connector	Type	Use
Type 1	AC	Single-phase AC charging in some markets.
Type 2	AC	Common AC charging connector; can support single/three phase.
GB/T	AC/DC variants	Chinese standard connector family.
CHAdeMO	DC	DC fast charging connector with communication.
CCS-1 / CCS-2	DC + AC combined	Combined Charging System for AC and DC pins.

Communication and standards

EV charging is not just power supply. Safe charging needs control pilot, proximity detection, charger-vehicle communication, insulation monitoring, protection, earthing, current limit and thermal supervision.

IEC 61851-1

General conductive charging system requirements and charging modes.

IEC 61851-24

Digital communication between DC EV charging station and electric vehicle.

IEC 62196-2

Requirements for AC plugs, socket-outlets and connectors.

IS / ARAI references

Indian charging infrastructure requirements should be checked during real installation and approval.

Field warning: EV charger installation is a high-current electrical job. Cable size, earthing, RCD/RCBO, surge protection, ventilation and standards compliance are not optional.

CALCULATION BANK

Important Formulas

Display

$$\text{Pixels} = \text{Horizontal pixels} \times \text{Vertical pixels}$$

$$\text{4K UHD} = 3840 \times 2160 = 8.29 \text{ million pixels}$$

Air conditioner

Cooling capacity in BTU/h = tonnage × 12000

Approx. tonnage = heat load / 12000

UPS and inverter

VA = W / PF

Backup time = V × Ah × η / W

Battery

Wh = V × Ah

Ah efficiency = Ah output / Ah input × 100%

Solar

Energy per day = Power × sunshine hours

No. of panels = Required power / panel rating

EV charging

Charging time = Battery kWh / Charger kW

3-phase power ≈ √3 × VL × IL × PF

SOLVED EXAMPLES

Worked Problems

Example 1: UPS rating

A computer load is 600 W at 0.8 power factor. Find UPS VA.

$$VA = 600 / 0.8 = 750\ VA$$

Select next standard rating, usually 1 kVA for margin.

Example 2: Battery backup

12 V, 100 Ah battery supplies 300 W load. Assume 80% efficiency.

$$\text{Backup} = 12 \times 100 \times 0.8 / 300 = 3.2\ h$$

Example 3: AC tonnage

A room requires 24000 BTU/h cooling. Find tonnage.

$$\text{Tonnage} = 24000 / 12000 = 2\ \text{ton}$$

Example 4: EV charging time

A 40 kWh EV battery is charged using 7.4 kW AC charger. Ignore losses.

$$\text{Charging time} = 40 / 7.4 = 5.4\ h$$

PRACTICE BANK

Expected Questions

Use these as revision prompts. Practice writing answers with diagram + working + comparison.

Module 1

Q1 Explain the evolution of electronic devices from vacuum tubes to integrated circuits.

Q2 Write short note on Moore's Law and its effect on consumer electronics.

Q3 Compare microprocessor and microcontroller.

Q4 Explain the role of microcontrollers in home appliances.

Q5 Draw a block diagram of a microcontroller-based consumer device.

Module 2

Q6 Explain construction and working of microphone and loudspeaker.

Q7 Explain 2.1 and 5.1 home theatre systems.

Q8 Compare LCD and LED display systems.

Q9 Compare 1080p, 4K and 8K resolution.

Q10 Compare VGA, AV, S-Video, HDMI and optical ports.

Module 3

Q11 Explain construction and working of washing machine.

Q12 Explain microwave oven and induction stove working principles.

Q13 Compare inverter and non-inverter AC. Explain tonnage.

Q14 Draw and explain on-grid and off-grid solar systems.

Q15 Compare online UPS, offline UPS and inverter.

Q16 Explain battery ratings, Ah efficiency, Wh efficiency and VRLA battery maintenance.

Module 4

Q17 Draw simple EV schematic and explain each block.

Q18 Compare slow charger and fast charger.

Q19 Compare AC charging and DC charging.

Q20 Explain EV charging Mode 2, Mode 3 and Mode 4.

Q21 List AC and DC charging connectors and applications.

Q22 Explain the importance of IEC 61851 and IEC 62196 standards.

OFFICIAL-PATTERN PRACTICE

Model Question Paper

CONTEMPORARY ELECTRONICS - 6042B
Time: 3 Hours   Maximum Marks: 75

PART A - Answer all questions. Each carries 2 marks.
1. Define consumer electronics.
2. What is Moore's Law?
3. State any two advantages of microcontrollers in appliances.
4. What is the function of a microphone?
5. Name any two video ports.
6. What is meant by 4K resolution?
7. Define UPS.
8. What is battery Ah rating?
9. What is EVSE?
10. Name any two DC fast charging connectors.

PART B - Answer any five. Each carries 5 marks.
11. Compare vacuum tube, transistor and IC.
12. Compare microprocessor and microcontroller.
13. Explain working of loudspeaker with neat points.
14. Compare LCD and LED displays.
15. Explain working principle of induction stove.
16. Compare online UPS and offline UPS.
17. Explain AC and DC EV charging.
18. Calculate backup time for a 24 V, 100 Ah battery supplying 500 W load at 80% efficiency.

PART C - Answer any five. Each carries 8 marks.
19. Draw and explain a microcontroller-based consumer electronics system.
20. Explain AM/FM receiver block diagram in an entertainment system.
21. Explain 2.1 and 5.1 home theatre systems with speaker arrangement.
22. Draw and explain off-grid solar power system.
23. Explain lead-acid and VRLA battery maintenance and safety.
24. Draw and explain simple electric vehicle schematic.

25. Explain EV charging modes 2, 3 and 4 with applications.
26. Explain charger connector selection: Type 1, Type 2, GB/T, CHAdeMO, CCS-1 and CCS-2.

ANSWER KEY

Model Paper Answer Hints

Part A answers

1. Consumer electronics are electronic products used in daily life for communication, entertainment, comfort and home automation.
2. Moore's Law states that transistor count in ICs tends to increase rapidly, improving performance and reducing cost per function.
3. Low cost, low power, compact size, reliable control, built-in timers/ADC/I/O.
4. Microphone converts sound energy into electrical signal.
5. VGA, AV, S-Video, HDMI, optical port.
6. 4K UHD is 3840 × 2160 pixel resolution.
7. UPS provides uninterrupted power backup during mains failure.
8. Ah rating expresses current capacity over time.
9. EVSE is Electric Vehicle Supply Equipment used for safe EV charging.
10. CHAdeMO, CCS-1, CCS-2, GB/T.

Part B / C marking points

- ✓ For comparison answers, use table format.
- ✓ For appliances, include construction, working sequence, control board and safety.
- ✓ For UPS/battery questions, include formulas and practical limitation.
- ✓ For EV charging, include AC/DC difference, mode, connector and communication.
- ✓ For diagrams, label every block and show signal/power direction.

Answer for Q18

$$\text{Backup} = V \times Ah \times \eta / W = 24 \times 100 \times 0.8 / 500 = 3.84 \text{ h}$$

References

- ✓ Thomas L Floyd, Electronic Devices, 10th Edition, Pearson Education Asia.

- ✓ Philp Hoff, Consumer Electronics for Engineers, Cambridge University Press.
- ✓ Dennis C Brewer, Home Automation, Que Publishing.
- ✓ Everyday Electronics course material provided by SITTTR.
- ✓ Tata Elxsi overview on EV charging infrastructure listed in syllabus resources.